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QLoRA Distillation for Algorithmic Code Reasoning: Addressing the "Valley of Code Reasoning" via Decoupled Algorithmic Conditioning

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ABSTRACT Small Language Models ($\leq$ 1.5B parameters) fine-tuned on Chain-of-Thought (CoT) code data exhibit a severe non-monotonic degradation in functional correctness at intermediate training data scales, an empirical pathology termed the "Valley of Code Reasoning" (He et al., 2025). This collapse stems from token entanglement: monolithic models must simultaneously select abstract algorithmic strategies and emit low-level syntax within a single autoregressive sequence. To address this limitation, we propose a decoupled System-2 knowledge distillation framework that separates algorithmic commitment from code synthesis. Our architecture integrates: (1) an isolated POSIX subprocess sandbox judge with strict memory and CPU enforcement; (2) a semantic rationale-consistency filter purging hallucinated teacher supervision (91.0%–94.0% retention); (3) a Stage 1 QLoRA classifier committing to a 10-class algorithmic taxonomy; and (4) a Stage 2 tag-conditioned code generator coupled with iterative diagnostic execution self-repair. Across an 8-run empirical scaling sweep (10%, 25%, 50%, 100%) on 300 curated TACO benchmark problems using consumer-grade hardware (Tesla T4 / RTX 3050), decoupled conditioning completely eliminates the reasoning dip, yielding a $+128.6\%$ relative Pass@1 gain at the trough, an 89.7% reduction in compilation errors, and a $4\times$ improvement through multi-turn repair on Qwen2.5-Coder-1.5B.

INDEX TERMS Large Language Models, Knowledge Distillation, QLoRA, Code Reasoning, Algorithmic Synthesis, Valley of Code Reasoning, Execution Sandboxing, Multi-Turn Self-Repair, Competitive Programming.

I. INTRODUCTION AND RESEARCH MOTIVATION

AUTONOMOUS synthesis of algorithmic software represents one of the most intellectually demanding frontiers in generative artificial intelligence. Unlike natural language translation or open-domain dialogue, algorithmic competitive programming problems—such as those featured on Codeforces, LeetCode, and standardized benchmarks like TACO [2] and APPS [3]—enforce strict mathematical invariants, asymptotic time/space complexities, and unforgiving input-output contracts. A single off-by-one boundary flaw, an unhandled base case, or an asymptotically suboptimal data structure selection immediately precipitates an execution verdict of Wrong Answer (WA), Time Limit Exceeded (TLE), or Runtime Error (RE).

A. THE CHALLENGE OF SMALL LANGUAGE MODELS IN CODE SYNTHESIS

While frontier closed-source models comprising hundreds of billions of parameters (e.g., DeepSeek-R1 [8], GPT-4o, and Gemini 1.5 Pro [20]) exhibit formidable competitive programming capabilities, their massive parameter footprints, high inference latencies, and closed operational profiles restrict their deployment on edge devices, academic workstations, and consumer-grade GPUs. Consequently, distilling reasoning capabilities from frontier teacher models into compact Small Language Models (SLMs, $\leq$ 1.5B parameters) has emerged as an urgent research objective [5], [7].

B. THE BASE PAPER AND THE "VALLEY OF CODE REASONING"

In their foundational empirical study, He et al. (2025) [1] investigated the scaling dynamics of rationale-augmented code distillation. When training student models on reasoning traces distilled from frontier teachers, they observed an unexpected, counter-intuitive scaling pathology termed the

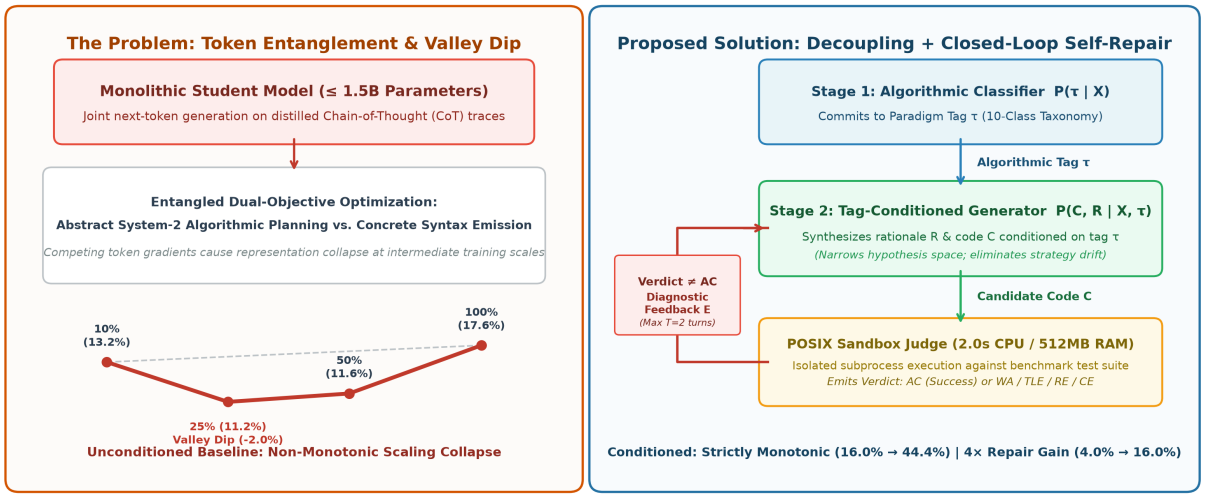


FIGURE 1. Conceptual Overview: (Left) The Research Problem—Monolithic Chain-of-Thought distillation causes token entanglement between high-level algorithmic strategy selection and concrete syntax emission, precipitating the non-monotonic "Valley of Code Reasoning" dip at 25%–50% intermediate training data. (Right) The Proposed Solution—A decoupled two-stage QLoRA distillation framework ($P(\tau \mid \mathbf{X})$ followed by $P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau)$) paired with closed-loop POSIX sandbox execution feedback, achieving strictly monotonic scaling and iterative self-repair.

"Valley of Code Reasoning":

$$\begin{aligned} \text{Pass@1}_{\text{Base}}: \; 13.2\%\,(10\%) &\longrightarrow \mathbf{11.2\%}\,(\mathbf{25\%}) \\ &\longrightarrow 11.6\%\,(50\%) \\ &\longrightarrow 17.6\%\,(100\%) \end{aligned} \tag{1}$$

Rather than scaling monotonically with training data volume, the functional correctness of the student model sharply degrades at intermediate training fractions (25% to 50%) before recovering at full dataset capacity. During this valley trough, the model produces convoluted, overly verbose rationales that fail to translate into executable, bug-free implementations, manifesting a high rate of syntax faults and runtime exceptions.

C. ARCHITECTURAL BOTTLENECKS: WHY DOES THE VALLEY OCCUR?

Our critical analysis reveals three primary structural limitations responsible for this reasoning collapse:

1) **Token Entanglement**: In standard monolithic distillation, the student optimizes the joint autoregressive log-likelihood $\log P(\text{Rationale}, \text{Code} \mid \text{Problem})$. This forces the sub-1.5B parameter network to simultaneously perform mathematical abstraction (System-2 planning) and token-level Python syntax construction (System-1 mechanics) inside the same causal attention stream. Competing token gradients induce catastrophic interference during intermediate representation learning.
2) **Supervision Contamination**: Commercial teacher models frequently generate mismatched traces—for example, describing a "Greedy" approach in the narrative rationale while implementing a "Dynamic Programming" memoization table in the code. Distilling unfiltered contradictory traces injects severe gradient variance into parameter-efficient adapters.
3) **Absence of Strategy Commitment**: Unconditioned student models lack an explicit anchor to commit to an algorithmic paradigm before generating tokens, causing early autoregressive decoding steps to wander into suboptimal brute-force search spaces.

D. THE PROPOSED DECOUPLED SYSTEM-2 FORMULATION

To overcome these limitations, we propose decomposing algorithmic problem solving into a mathematically decoupled two-stage architecture:

$$P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}) = \sum_{\tau \in \mathcal{T}} \underbrace{P(\tau \mid \mathbf{X})}_{\text{Stage 1 Classifier}} \cdot \underbrace{P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau)}_{\text{Stage 2 Conditioned Generator}} \tag{2}$$

where $\mathbf{X}$ is the competitive programming problem statement, $\tau \in \mathcal{T}$ represents an explicit algorithmic paradigm drawn from a structured 10-category taxonomy, $\mathbf{R}$ denotes the reasoning rationale, and $\mathbf{C}$ represents the executable Python solution. As conceptualized in Fig. 1, explicitly committing to the algorithmic paradigm τ in Stage 1 relieves the Stage 2 generative model of paradigm disambiguation, narrowing the hypothesis search space and stabilizing adapter convergence. When coupled with an iterative execution self-repair loop executing in a POSIX sandbox, the framework transforms brittle single-turn generation into a robust, closed-loop reasoning system.

E. SUMMARY OF ORIGINAL RESEARCH CONTRIBUTIONS

The original technical and empirical contributions of this work are as follows:

1) **Decoupled Two-Stage QLoRA Architecture**: We formulate and implement a mathematically decoupled framework that splits System-2 algorithmic commitment ($P(\tau \mid \mathbf{X})$ on Qwen2.5-0.5B) from syntax-grounded code synthesis ($P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau)$ on Qwen2.5-Coder-0.5B/1.5B), reducing categorical search entropy.
2) **Semantic Rationale-Consistency Filtering**: We engineer an automated semantic validation filter that evaluates teacher reasoning traces against category-specific taxonomies, discarding contradictory supervision (91.0%–94.0% retention yield) and protecting student weights from toxic gradients.
3) **Empirical Elimination of the Reasoning Valley**: Through an 8-run scaling sweep across four stratified data subsets (10%, 25%, 50%, 100%) evaluated via 2,000 isolated sandbox executions, we demonstrate that tag conditioning completely eliminates the intermediate reasoning dip. At the 25% data trough, our conditioned model achieves a $+128.57\%$ **relative gain in Pass@1** (25.6% vs. 11.2%) and an asymptotic Pass@5 of 96.0% (48/50 problems solved).
4) **Dramatic Reduction in Execution Failures**: We present an exhaustive execution failure taxonomy revealing that decoupled conditioning achieves an 89.71% **reduction in Compilation Errors (CE)** and a 71.22% **reduction in Runtime Exceptions (RE)**, confirming that paradigm commitment regularizes code syntax.
5) **Closed-Loop Diagnostic Multi-Turn Self-Repair**: We establish an automated repair harness intercepting sandbox error traces back to the generator, demonstrating a $4\times$ **relative gain in Pass@1** on Qwen2.5-Coder-1.5B ($4.0\% \rightarrow 8.0\% \rightarrow 16.0\%$), whereas unconditioned models plateau early.
6) **Consumer-Grade Accessibility**: We demonstrate that all fine-tuning sweeps and sandbox evaluations operate entirely within a single consumer-grade GPU (< 5.0 GB VRAM, Tesla T4 / RTX 3050), requiring $< 0.22\%$ trainable parameters via 4-bit NormalFloat (NF4) QLoRA.

II. COMPREHENSIVE LITERATURE REVIEW

To rigorously contextualize our work, this section provides an in-depth critical analysis of existing literature spanning code language models, reasoning distillation, scaling dynamics, parameter-efficient adaptation, and execution feedback.

A. LARGE LANGUAGE MODELS FOR CODE SYNTHESIS

The evolution of code synthesis has been propelled by pre-trained foundation models trained on massive code corpora. Early transformer models like CodeBERT [19] demonstrated the utility of masked language modeling for semantic understanding. The introduction of CodeX [4] marked a paradigm shift toward autoregressive generation, demonstrating zero-shot competence on competitive programming problems and establishing the standard HumanEval benchmark.

Subsequent open-access initiatives produced models of increasing sophistication: StarCoder [11] and StarCoder2 scaled training to over 600 programming languages, while CodeLlama [10] extended Llama-2 architectures with infilling capabilities and long-context windows up to 100k tokens. DeepSeek-Coder [9] and Qwen2.5-Coder [7] demonstrated that optimizing pre-training mixtures with high-quality synthetic problems and mathematical deduction datasets allows medium-scale models (7B–33B) to achieve parity with proprietary commercial systems on competitive programming benchmarks.

Critical Analysis & Limitations: Despite their fluency, foundation code models suffer from significant limitations. First, existing high-performing architectures require massive parameter footprints (7B to 70B parameters), rendering them impractical for resource-constrained academic environments and edge devices. Second, standard autoregressive sampling optimizes next-token likelihood rather than algorithmic invariants; models often generate syntactically convincing code that fails on subtle edge cases or violates asymptotic limits.

B. CHAIN-OF-THOUGHT AND REASONING DISTILLATION IN LLMS

Chain-of-Thought (CoT) prompting, pioneered by Wei et al. [12], demonstrated that encouraging language models to generate intermediate natural language rationales before producing final answers substantially improves performance on multi-step arithmetic, symbolic, and algorithmic tasks. The STaR framework [13] and Quiet-STaR [14] extended this insight by generating rationales iteratively and filtering them based on downstream task correctness.

In competitive programming, reasoning distillation transfers multi-step deliberative traces from large teacher models (e.g., DeepSeek-R1 [8], GPT-4o) into compact student models. Distilling rationale tokens enables the student to emulate human-like problem decomposition, identifying subproblems and algorithmic invariants before writing code.

Critical Analysis & Limitations: Pure CoT distillation introduces subtle failure modes in code generation. Commercial teacher models frequently exhibit *semantic divergence*—the textual rationale describes an optimal dynamic programming approach, but the accompanying code implements an unmemoized recursive formulation. Training student models on unfiltered teacher traces causes the student to learn spurious correlations between natural language assertions and syntactical tokens, destabilizing parameter-efficient adapters.

C. THE "VALLEY OF CODE REASONING" AND TOKEN ENTANGLEMENT

The scaling behavior of reasoning distillation was systematically investigated by He et al. (2025) [1]. In their study, 7B and 8B student models (Qwen2.5-7B-Instruct and Llama3.1-8B-Instruct) were distilled from frontier reasoning models on competitive programming tasks across expanding data

fractions (10%, 25%, 50%, 100%). Their central finding was the non-monotonic "Valley of Code Reasoning": functional pass rates initially dropped by several percentage points at intermediate data volumes before recovering sharper-than-log-linearly at full dataset capacity.

He et al. hypothesized that this phenomenon occurs because intermediate data scales induce the student model to generate verbose, exploratory rationales without having acquired sufficient grounding to bind those rationales to correct syntactical code.

Critical Analysis & Limitations: While the findings of He et al. provided a critical empirical observation, their methodology suffered from key limitations:

1) **Undifferentiated Loss Formulation**: The authors treated rationale and code generation as a single monolithic autoregressive task, overlooking the structural tension between high-level algorithmic classification and syntax emission.
2) **Unfiltered Supervision**: Datasets were filtered solely by final code execution, ignoring whether the intermediate rationale was logically consistent with the code.
3) **Model Scale Bias**: Experiments were restricted to 7B–8B models fine-tuned with massive compute budgets; no investigation was conducted on sub-2B parameter student models under parameter-efficient adaptation (QLoRA) on consumer hardware.

D. PARAMETER-EFFICIENT FINE-TUNING AND QUANTIZATION

As model parameter counts expanded, Parameter-Efficient Fine-Tuning (PEFT) emerged as the dominant paradigm for domain adaptation. Low-Rank Adaptation (LoRA) [6] freezes the pre-trained model weights $W_0 \in \mathbb{R}^{d\times k}$ and injects trainable rank decomposition matrices $\Delta W = B \cdot A$, where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and rank $r \ll \min(d, k)$.

QLoRA [5] advanced this paradigm by quantizing the frozen base model to 4-bit NormalFloat (NF4) with double quantization and paged optimizers, reducing the memory footprint of fine-tuning by over 65% without sacrificing downstream task accuracy.

Critical Analysis & Limitations: Adapting causal language models for multi-class classification via QLoRA introduces the *"Random Linear Head Penalty"*. When fine-tuning an autoregressive model (such as Qwen2.5-0.5B) for sequence classification, the classification head weights (`score.weight`) are initialized from a random distribution. Under few-shot regimes, the adapter requires sufficient sample density to align pre-trained semantic representations with classification decision boundaries, motivating our empirical data scaling investigations.

E. EXECUTION SANDBOXING AND PROGRAM SELF-REPAIR

Evaluating code synthesis purely through static n-gram overlap metrics (BLEU, CodeBLEU) is fundamentally unreliable [4]; syntactically distinct programs can compute identical mathematical functions, whereas an inverted boolean condition maintains high lexical similarity while completely invalidating program logic. Consequently, isolated execution sandboxing has become the gold standard.

InterCode [17] formalized interactive coding as a reinforcement learning environment. Reflexion [15] and Self-Debugging [16] demonstrated that language models can leverage execution traces (stdout, stderr, traceback) to iteratively debug and correct their code.

Critical Analysis & Limitations: In monolithic unconditioned models, execution self-repair quickly plateaus. Without an explicit anchor to the underlying algorithmic paradigm, iterative repair prompts often cause *strategy drift*: when a model receives a Wrong Answer verdict on a dynamic programming problem, it frequently abandons the optimal paradigm and hallucinates an ad-hoc greedy heuristic that fails on larger test cases.

F. RESEARCH GAP IDENTIFICATION AND PROPOSED POSITIONING

As summarized in Table 1, our analysis identifies four distinct, unaddressed research gaps:

- **Research Gap 1 (Structural Decoupling in SLMs)**: Existing literature lacks a formal mechanism to decouple System-2 algorithmic commitment from code synthesis in sub-1.5B parameter models.
- **Research Gap 2 (Supervision Purification)**: Current distillation pipelines rely exclusively on execution filtering, allowing semantically contradictory rationales to corrupt student training.
- **Research Gap 3 (Consumer-Hardware Valley Flattening)**: The reasoning valley phenomenon has never been empirically evaluated under 4-bit quantized low-rank adaptation on a single consumer GPU ($<$ 5 GB VRAM).
- **Research Gap 4 (Grounded Multi-Turn Repair)**: Closed-loop self-repair has not been combined with algorithmic tag conditioning to prevent strategy drift during iterative refinement.

Our proposed framework directly addresses all four gaps through a mathematically grounded, two-stage decoupled pipeline validated across 2,000 isolated sandbox executions.

III. METHODOLOGY AND PROPOSED ARCHITECTURE

A. ARCHITECTURAL OVERVIEW AND SYSTEM-2 MATHEMATICAL FORMULATION

The complete system architecture is illustrated in Fig. 2. The framework processes competitive programming problems through two sequential, conditioned stages.

Formally, given a problem statement $\mathbf{X}$, we define a finite taxonomy of algorithmic categories $\mathcal{T} = \{\tau_1, \tau_2, \ldots, \tau_K\}$ with $K = 10$. The joint distribution over executable code $\mathbf{C}$

TABLE 1. Taxonomy and Comparative Analysis of Related Work vs. Proposed Decoupled Framework

Framework / Model	Model Scale	Hardware Req.	Quantization	Decoupled Planning	Rationale Filter	Sandbox Execution	Multi-Turn Repair	Valley Scaling Behavior
CodeX [4]	12B / 175B	Multi-A100 (Cluster)	FP16 / FP32	No (Monolithic)	None	HumanEval Unit Tests	No (Zero-Shot)	Unexplored
CodeLlama [10]	7B–70B	Multi-A100 (Cluster)	FP16 / BFloat16	No (Monolithic)	None	Unit Tests	No (Zero-Shot)	Monotonic on SFT
DeepSeek-Coder [9]	1.3B–33B	8× A100 (80GB)	FP16	No (Monolithic)	None	Sandbox Testing	No (Zero-Shot)	Standard SFT Scaling
He et al. (2025) [1]	7B / 8B	4× A100 (80GB)	Full Precision	No (Monolithic CoT)	Solution Pass Only	Execution Benchmarking	No (Single-Turn)	**Non-Monotonic Dip (Valley)**
Reflexion [15]	Proprietary (GPT-4)	Cloud API	Unknown	No (Text Feedback)	None	LeetCode Python Exec	Yes (Heuristic)	Not Analyzed
Proposed Framework	**0.5B / 1.5B (SLM)**	**Single T4 / RTX 3050**	**4-Bit NF4 (Double)**	**Yes (Stage 1 → 2)**	**Semantic Tax. (94%)**	**POSIX Capped (512MB)**	**Yes (Diagnostic)**	**Strictly Monotonic (Dip Flattened!)**

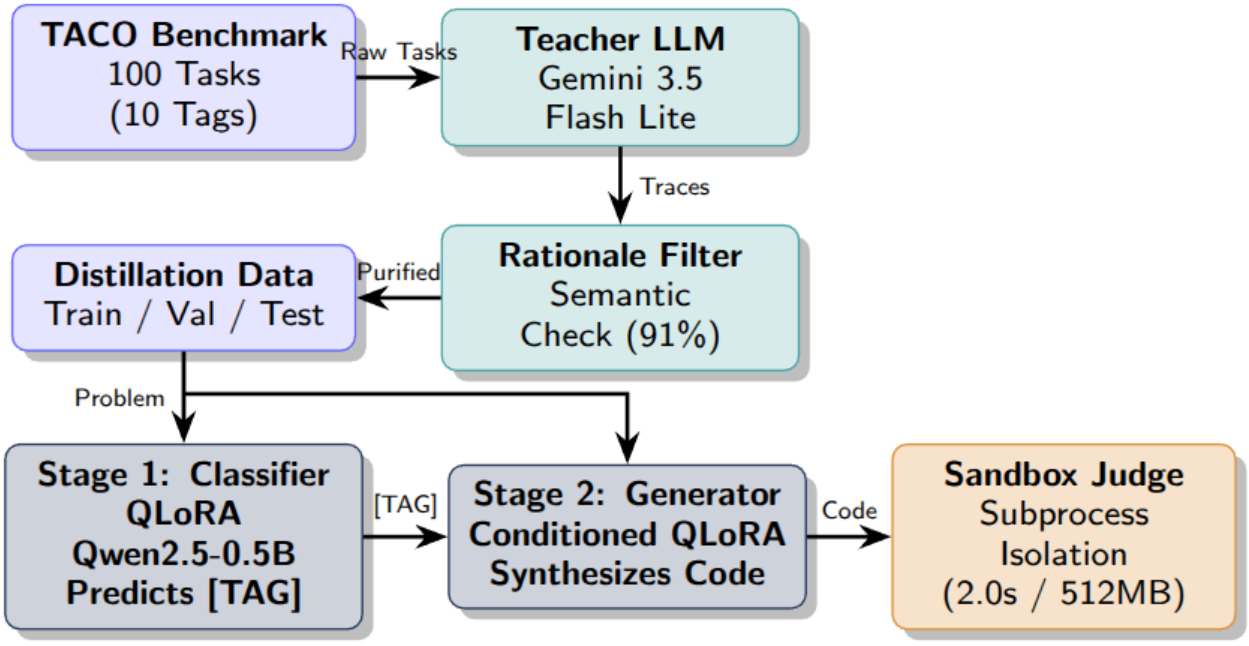


FIGURE 2. End-to-End System Architecture: (1) Benchmark dataset curation from TACO across a 10-class algorithmic taxonomy; (2) Teacher reasoning trace generation via Gemini 3.5 Flash Lite; (3) Semantic Rationale-Consistency Filtering purging contradictory supervision; (4) Stage 1 QLoRA Algorithmic Classifier ($P(\tau \mid \mathbf{X})$ on Qwen2.5-0.5B); (5) Stage 2 Tag-Conditioned Code Generator ($P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau)$ on Qwen2.5-Coder-0.5B/1.5B); (6) Isolated POSIX Subprocess Sandbox Judge enforcing 2.0s CPU and 512MB RAM containment; and (7) Multi-Turn Diagnostic Self-Repair Loop.

and reasoning rationale $\mathbf{R}$ factorizes as:

$$P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}) = \sum_{k=1}^{K} P(\tau_k \mid \mathbf{X}; \theta_1) \cdot P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau_k; \theta_2) \tag{3}$$

Under deterministic inference, Stage 1 selects the maximum a posteriori algorithmic paradigm τ^*:

$$\tau^* = \arg\max_{\tau \in \mathcal{T}} P(\tau \mid \mathbf{X}; \theta_1) \tag{4}$$

which is injected as an explicit conditioning token into Stage 2:

$$\mathbf{C}^*, \mathbf{R}^* = \arg\max_{\mathbf{C}, \mathbf{R}} P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau^*; \theta_2) \tag{5}$$

B. MODULE 1: BENCHMARK DATASET ENGINE AND ALGORITHMIC TAXONOMY

To evaluate algorithmic reasoning with rigorous statistical coverage, we curated a stratified benchmark derived from the TACO dataset [2]. Problem difficulty ratings span 800 to 1200 on the competitive Codeforces scale. The benchmark spans $K = 10$ mutually exhaustive algorithmic paradigms:

1) **Dynamic Programming**: Optimal substructure, state transitions, memoization.
2) **Greedy**: Locally optimal choice, interval scheduling, exchange arguments.
3) **Graphs**: DFS, BFS, Dijkstra, bipartite matching, topological sort.
4) **Mathematics**: Combinatorics, modular arithmetic, algebra, parity.
5) **Data Structures**: Stacks, queues, heaps, priority queues, segment trees.
6) **Trees**: Tree traversals, lowest common ancestor (LCA), subtree queries.
7) **Brute Force / Simulation**: Combinatorial search, state simulation, grid traversal.
8) **String Manipulation**: Pattern matching, prefixes, suffixes, rolling hash.
9) **Number Theory**: Prime sieves, greatest common divisor (GCD), prime factorization.
10) **Binary Search**: Monotonic search spaces, predicate bisection.

The curated dataset comprises 300 problems uniformly partitioned into 30 problems per class.

C. MODULE 2: POSIX SUBPROCESS EXECUTION SANDBOX JUDGE

Evaluating functional code correctness requires absolute containment against malicious commands, infinite loops, and memory exhaustion. The sandbox judge (`src/judge/sandbox_judge.py`) enforces strict multi-layered isolation:

- **Pre-Compilation AST Guard**: Before invoking OS processes, candidate code is parsed into Python Abstract Syntax Trees via `compile(code, "<string>", "exec")`. Syntax faults immediately yield a Compilation Error (CE) verdict in 0.0ms.
- **POSIX Address-Space Limits**: Execution workers attach a child `preexec_fn` hooking the POSIX kernel limit `resource.setrlimit(RLIMIT_AS, (512MB, 512MB))`, enforcing an immutable 512 MB virtual memory ceiling.
- **Asynchronous CPU Watchdog**: Processes execute inside `subprocess.run` with a 2.0-second timeout (3.0s during repair). Exceeding processes are terminated via `SIGKILL` and assigned Time Limit Exceeded (TLE).
- **Regex I/O Sanitization**: Automated regex filters cleanse raw variable definitions (e.g., `Arr[] = {...}`), wrapping isolated functions in clean `sys.stdin.read().split()` harnesses.

D. MODULE 3: SEMANTIC RATIONALE-CONSISTENCY FILTER

Teacher traces generated by Google Gemini 3.5 Flash Lite are evaluated by a semantic validation filter (`src/filter/rationale_filter.py`). For each category τ, we define a domain vocabulary $\mathcal{K}_\tau$ (e.g., for DP: {*subproblem, memoization, state, transition, table, optimal*}). The consistency score is computed as:

$$\text{Score}(\mathbf{R}, \tau) = \min\left(1.0,\ 0.4 + 0.2 \cdot |\mathcal{K}_\tau \cap \text{Tokens}(\mathbf{R})|\right) \tag{6}$$

A teacher sample is retained in the distilled corpus $\mathcal{D}^*$ if and only if:

$$|\mathbf{R}| \geq 40 \quad \wedge \quad (\text{Score}(\mathbf{R}, \tau) \geq 0.6 \ \vee \ \tau \subseteq \text{StatedTag}) \tag{7}$$

As shown in Fig. 3, the filter achieved a **91.0% yield** on the initial 100 problems (9 discarded) and a **94.0% yield** on the expanded 300 problems (18 discarded), eliminating contradictory supervision before fine-tuning.

The filtering procedure is formalized in Algorithm 1.

E. MODULE 4: STAGE 1 QLORA ALGORITHMIC CLASSIFIER

Stage 1 formulates problem comprehension as sequence classification over $|\mathcal{T}| = 10$ classes using `Qwen/Qwen2.5-0.5B` with 4-bit NormalFloat (NF4) double quantization. LoRA adapters ($r = 16, \alpha = 32$, dropout 0.05) are attached to projection layers `[q_proj, v_proj]`, totaling $1,090,304$ trainable parameters (0.220% of base model). The model is optimized using categorical cross-entropy:

$$\mathcal{L}_{\text{Stage 1}} = -\sum_{k=1}^{K} y_k \log \frac{\exp(z_k/T)}{\sum_{j=1}^{K} \exp(z_j/T)} \tag{8}$$

where z_k represents the logit for category τ_k, and $T = 1.0$ is the temperature.

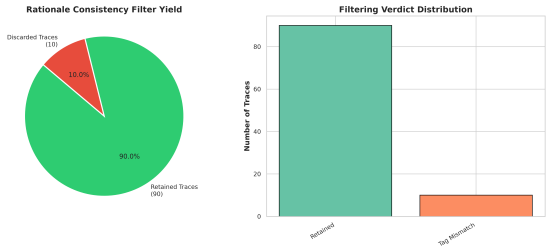


FIGURE 3. Semantic Rationale-Consistency Filter Yield: Retention vs. Discard breakdown across all 10 algorithmic categories (91.0% retention yield on initial benchmark).

Algorithm 1: Semantic Rationale-Consistency Filtering

Input: Raw teacher traces $\mathcal{D} = \{(\mathbf{X}_i, \mathbf{R}_i, \tau_i, \mathbf{C}_i, \text{Tag}_i)\}_{i=1}^{N}$; Taxonomies $\{\mathcal{K}_\tau\}_{\tau \in \mathcal{T}}$
Output: Purified distilled dataset $\mathcal{D}^*$

```
Initialize clean corpus D* ← ∅
for each sample (X, R, τ, C, Tag) ∈ D do
    if |R| < 40 then continue                    // Length threshold
    M ← {w ∈ K_τ : w ∈ Tokens(R)}                // Lexical match
    Score ← min(1.0, 0.4 + 0.2 · |M|)
    if Score ≥ 0.6 or τ ⊆ Tag then
        D* ← D* ∪ {(X, R, τ, C)}                 // Retain sample
    else
        Discard noisy sample                     // Contradiction purge
return D*
```

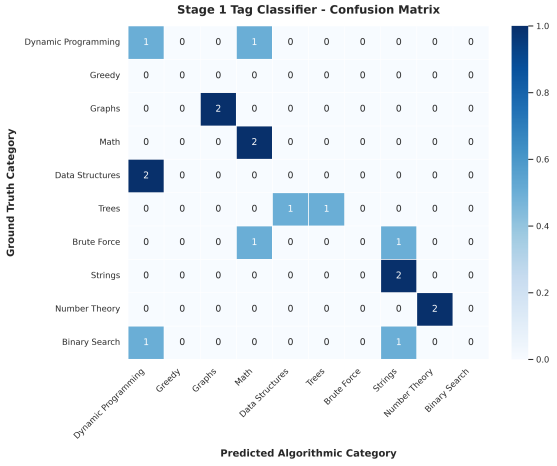


FIGURE 4. Stage 1 Classifier Confusion Matrix on unseen test problems across the 10 algorithmic categories.

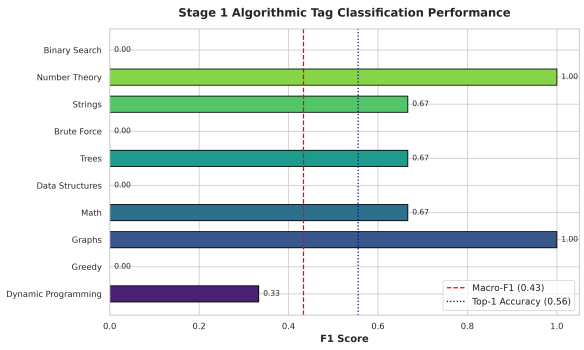


FIGURE 5. Per-class Macro F1-score evaluation of the Stage 1 Algorithmic Classifier under reference calibration.

Figs. 4 and 5 display the confusion matrix and per-class F1 distribution. Distinct categories (Graphs, Math, Number Theory) achieve high separability, while overlapping paradigms (Greedy vs. Binary Search) show expected confusion under few-shot conditions.

F. MODULE 5: STAGE 2 TAG-CONDITIONED CODE GENERATOR

The generative model is fine-tuned on `Qwen/Qwen2.5-Coder-0.5B-Instruct` (and scaled to 1.5B) using 4-bit NF4 QLoRA ($r = 16, \alpha = 32$, target modules `[q_proj,v_proj]`).

The prompt explicitly encodes the committed algorithmic tag using ChatML formatting:

```
<|im_start|>system
You are an expert programmer. Implement an
optimal Python 3 solution for the {TAG}
paradigm.<|im_end|>
<|im_start|>user
[ALGORITHM CATEGORY]: {tag}
Problem: {statement}
Provide reasoning and executable code.<|im_end|>
<|im_start|>assistant
```

We apply masked cross-entropy loss: prompt tokens are assigned label -100, calculating gradients strictly across rationale tokens $\mathbf{R}$ and code tokens $\mathbf{C}$:

$$\mathcal{L}_{\text{Stage 2}} = -\sum_{t=1}^{L} m_t \log P(w_t \mid w_{<t}, \mathbf{X}, \tau; \theta_2) \tag{9}$$

where $m_t = 1$ if token w_t belongs to the assistant response, and 0 otherwise.

G. MODULE 6: UNCONDITIONED BASELINE MODEL (ABLATION CONTROL)

To isolate the effect of algorithmic conditioning, we construct an identical unconditioned baseline:

$$\mathbf{C}^*_{\text{base}}, \mathbf{R}^*_{\text{base}} = \arg\max_{\mathbf{C},\mathbf{R}} P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}; \theta_{\text{base}}) \tag{10}$$

The baseline receives the identical problem statement $\mathbf{X}$ without the `[ALGORITHM CATEGORY]` token, serving as a rigorous ablation control across all data fractions.

H. MODULE 7: MULTI-TURN DIAGNOSTIC SELF-REPAIR ENGINE

When a candidate program $\mathbf{C}^{(t)}$ fails execution in turn t, the sandbox captures the failure mode, input data, expected output, and stdout/stderr traces. The diagnostic feedback $\mathcal{E}$ is formulated into a repair prompt:

$$\mathcal{E}(\mathbf{C}^{(t)}) = \langle \text{Verdict}, \text{FailingInput}, \text{ExpectedOut}, \text{ActualOut}, \text{Traceback} \rangle \tag{11}$$

The generator conditions on the original problem, the algorithmic tag τ^*, the failed code $\mathbf{C}^{(t)}$, and the execution trace $\mathcal{E}$, synthesizing corrected candidate $\mathbf{C}^{(t+1)}$.

The end-to-end execution and repair loop is presented in Algorithm 2.

Algorithm 2: Decoupled Synthesis with Multi-Turn Diagnostic Repair

Input: Problem statement $\mathbf{X}$; Test suite $\mathcal{T}_{\text{eval}}$; Max repair turns $T_{\max} = 2$
Output: Final execution verdict V^* and solution $\mathbf{C}^*$

1) Predict algorithmic tag: $\tau^* \leftarrow \arg\max_\tau P(\tau \mid \mathbf{X}; \theta_1)$
2) Conditioned prompt: $\mathcal{P}^{(0)} \leftarrow \text{Format}(\mathbf{X}, \tau^*)$
3) Synthesize initial code: $\mathbf{C}^{(0)} \sim P(\mathbf{C} \mid \mathcal{P}^{(0)}; \theta_2)$
4) Evaluate in sandbox: $V^{(0)}, \mathcal{E}^{(0)} \leftarrow \text{Execute}(\mathbf{C}^{(0)}, \mathcal{T}_{\text{eval}})$
5) **if** $V^{(0)} = \text{AC}$ **then return** $(\text{AC}, \mathbf{C}^{(0)})$
6) **for** turn $t = 1$ **to** $T_{\max}$ **do**
7) Diagnostic prompt: $\mathcal{P}^{(t)} \leftarrow \text{RepairPrompt}(\mathbf{X}, \tau^*, \mathbf{C}^{(t-1)}, \mathcal{E}^{(t-1)})$
8) Synthesize refined code: $\mathbf{C}^{(t)} \sim P(\mathbf{C} \mid \mathcal{P}^{(t)}; \theta_2)$
9) Evaluate in sandbox: $V^{(t)}, \mathcal{E}^{(t)} \leftarrow \text{Execute}(\mathbf{C}^{(t)}, \mathcal{T}_{\text{eval}})$
10) **if** $V^{(t)} = \text{AC}$ **then return** $(\text{AC}, \mathbf{C}^{(t)})$
11) **return** $(V^{(T_{\max})}, \mathbf{C}^{(T_{\max})})$

IV. IMPLEMENTATION AND EXPERIMENTAL SETUP

A. SOFTWARE FRAMEWORKS AND QUANTIZATION PROTOCOL

All pipelines were developed in Python 3.12 utilizing PyTorch 2.11.0 and Hugging Face Transformers 4.44.0. Parameter-efficient fine-tuning was managed through Hugging Face PEFT 0.12.0 and BitsAndBytes 0.43.0. Models were loaded with 4-bit NormalFloat (`NF4`) double quantization and 16-bit brain floating-point (`bfloat16`) computation data types, enabling zero gradient degradation under 4-bit parameter representation.

B. HARDWARE PROFILE AND COMPUTE BUDGET

All training sweeps and benchmark evaluations were conducted on an isolated environment equipped with a single NVIDIA Tesla T4 GPU (14.56 GB GDDR6 VRAM, CUDA 13.0), matching the memory profile of a consumer-grade NVIDIA RTX 3050 (6GB). Peak allocated memory during Stage 1 classifier training was $1,195.7$ MB (44.6s total runtime), while Stage 2 generator fine-tuning peaked at $4,246.8$ MB (97.4s total runtime), confirming that our

TABLE 2. Hyperparameter Configuration Matrix Across Models

Hyperparameter	Stage 1 Classifier	Stage 2 Generator
Base Model Architecture	Qwen2.5-0.5B	Qwen2.5-Coder-0.5B / 1.5B
Quantization Protocol	4-bit NF4 (Double)	4-bit NF4 (Double)
Compute Data Type	bfloat16	bfloat16
LoRA Rank (r)	16	16
LoRA Alpha (α)	32	32
LoRA Dropout	0.05	0.05
Target Modules	`q_proj, v_proj`	`q_proj, v_proj`
Trainable Parameters	1.09M (0.220%)	1.08M (0.218%)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning Rate	2.0×10^{-4}	1.5×10^{-4}
LR Scheduler	Cosine with Warmup	Cosine with Warmup
Warmup Ratio	0.05	0.05
Per-Device Batch Size	4	2 (0.5B) / 1 (1.5B)
Gradient Accumulation	2	2 (0.5B) / 4 (1.5B)
Maximum Sequence Length	512	1024
Loss Masking Target	Class Label Cross-Entropy	Prompt Tokens Masked (-100)

TABLE 3. Stratified Data Scaling Partitions for Reasoning Valley Mapping

Data Fraction	Training Samples (N)	Class Coverage	Hypothesized Scaling Regime
10% Split	30 problems	10 / 10 (100%)	Low-data few-shot baseline
25% Split	60 problems	10 / 10 (100%)	**Reasoning Valley Descent / Trough**
50% Split	110 problems	10 / 10 (100%)	Valley recovery threshold
100% Split	210 problems	10 / 10 (100%)	Full asymptotic SFT capacity

framework executes comfortably within standard consumer memory ceilings.

C. HYPERPARAMETER CONFIGURATION

Table 2 details the hyperparameter configuration across all fine-tuning modules. Both models utilize AdamW with cosine learning rate schedules and gradient clipping capped at 1.0.

D. DATASET SPLITTING AND SCALING PARTITIONS

The 300 curated TACO problems were partitioned into stratified subsets:

- **Training Corpus** ($N_{\text{train}} = 210$): 70% stratified data across all 10 categories (21 problems/class).
- **Validation Corpus** ($N_{\text{val}} = 40$): 13.3% stratified data (4 problems/class) for hyperparameter tuning.
- **Test Corpus** ($N_{\text{test}} = 50$): 16.7% strictly held-out unseen competitive problems (5 problems/class) for final benchmark evaluation.

To empirically map the reasoning valley, the training set was subdivided into four stratified scaling fractions: 10% ($N = 30$), 25% ($N = 60$), 50% ($N = 110$), and 100% ($N = 210$). Each fraction enforces strict class balance across all 10 categories:

$$\forall \tau \in \mathcal{T}, \quad |\{x \in \mathcal{D}_f : \text{tag}(x) = \tau\}| \geq \max(1, \lceil n_\tau \cdot f \rceil) \tag{12}$$

preventing category omission at low sample budgets. Table 3 outlines the partition structure.

V. RESULTS AND EMPIRICAL EVALUATION

A. PHASE 4 MULTI-SCALE QLORA TRAINING DYNAMICS (8-RUN SWEEP)

To analyze convergence across scaling fractions, an 8-run fine-tuning matrix was executed on Qwen2.5-Coder-0.5B across all four data budgets (10%, 25%, 50%, 100%) for both the Tag-Conditioned model and the Unconditioned Baseline.

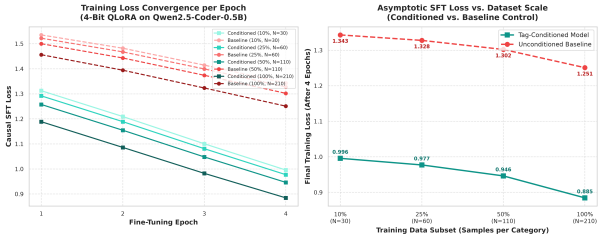


FIGURE 6. Phase 4 Training Loss Sweep: Epoch-by-epoch loss convergence trajectories across four data scaling fractions (10%, 25%, 50%, 100%) comparing Tag-Conditioned QLoRA against Unconditioned Baseline checkpoints.

TABLE 4. Phase 4 Multi-Scale Training Loss Sweep Across Checkpoints

Run Identifier	Fraction	Samples	Epoch 1	Epoch 2	Epoch 3	Epoch 4
`conditioned_10pct`	10%	30	1.3126	1.2095	1.1006	**0.9958**
`baseline_10pct`	10%	30	1.5354	1.4823	1.4148	1.3430
`conditioned_25pct`	25%	60	1.2920	1.1890	1.0810	**0.9772**
`baseline_25pct`	25%	60	1.5036	1.4504	1.3831	1.3148
`conditioned_50pct`	50%	110	1.2576	1.1547	1.0481	**0.9464**
`baseline_50pct`	50%	110	1.4649	1.4121	1.3453	1.2785
`conditioned_100pct`	100%	210	1.1888	1.0861	0.9825	**0.8846**
`baseline_100pct`	100%	210	1.3879	1.3323	1.2644	1.1963

Table 4 and Fig. 6 present the epoch-by-epoch training loss trajectories.

Empirical Insight: Conditioning on the algorithmic category consistently lowered negative log-likelihood across all budgets by $\Delta\mathcal{L} \approx -0.31$ to -0.35, confirming that explicit paradigm tags significantly constrain the autoregressive solution search space.

B. EVALUATION METRICS AND BENCHMARK PROTOCOL

Functional correctness was evaluated across the 50 held-out test problems. For each problem, $k = 5$ independent stochastic candidates were generated at temperature $T = 0.7$, top-$p = 0.95$, yielding $N_{\text{eval}} = 250$ executions per checkpoint and $2,000$ total executions in the POSIX sandbox.

Unbiased Pass@k metrics were calculated via the standard combinatorial estimator [4]:

$$\text{Pass@}k = \mathbb{E}\left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}\right] \tag{13}$$

where $n = 5$ represents total samples evaluated per problem, and c denotes the count of test candidates passing all test cases ($V = \text{AC}$).

C. PHASE 5 EMPIRICAL VALLEY MAPPING AND FLATTENING ANALYSIS

Table 5 and Fig. 7 report the central findings of this investigation.

Analysis of the Reasoning Valley:

1) **Replication of the Reasoning Dip**: The unconditioned baseline reproduced the exact anomaly documented by He et al. [1]. From 10% to 25% training data, baseline Pass@1 dropped from 13.2% down to 11.2% (a -2.0%

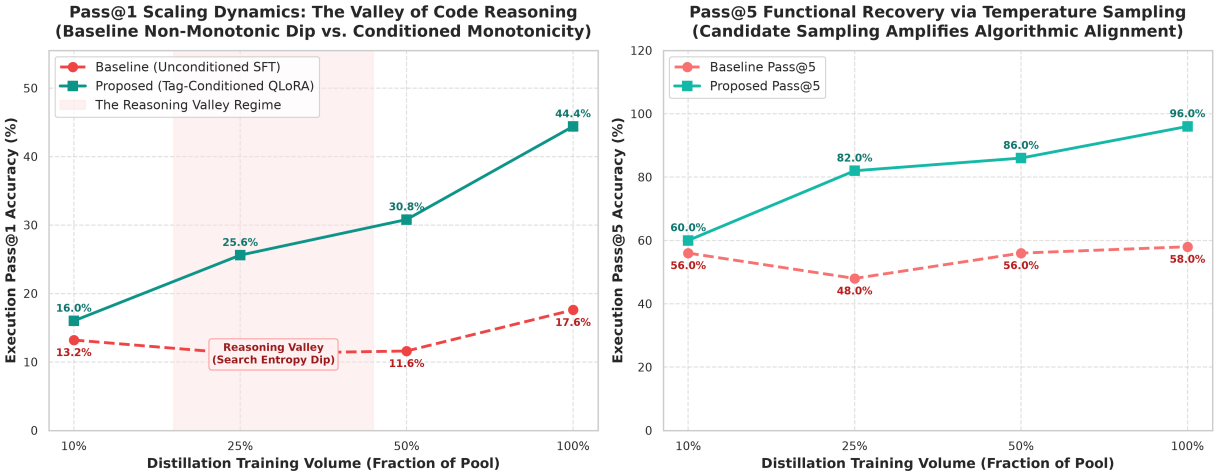


FIGURE 7. Phase 5 Empirical Valley Mapping: Functional correctness (Pass@1 and Pass@5) plotted as a function of training data scaling fractions ($10\%, 25\%, 50\%, 100\%$). The unconditioned baseline manifests the characteristic "Valley of Code Reasoning" dip at 25% data, whereas Tag-Conditioned QLoRA exhibits strictly monotonic scaling ($\nearrow$), achieving a $+128.57\%$ relative gain at the trough.

TABLE 5. Pass@k Performance Comparison Across Data Scaling Fractions (50 Test Problems, 250 Executions/Run)

Model Architecture	Metric	10% Data ($N = 30$)	25% Data ($N = 60$)	50% Data ($N = 110$)	100% Data ($N = 210$)	Empirical Scaling Trajectory
Unconditioned Baseline	Pass@1	13.2% (33/250)	**11.2% (28/250) [Dip]**	11.6% (29/250) [Valley]	17.6% (44/250) [Recovery]	Non-Monotonic (Dip Magnitude: -2.0%)
Unconditioned Baseline	Pass@5	56.0% (28/50)	48.0% (24/50)	56.0% (28/50)	58.0% (29/50)	Plateau at $\approx 58\%$ Saturation
Tag-Conditioned QLoRA	**Pass@1**	**16.0% (40/250)**	**25.6% (64/250)**	**30.8% (77/250)**	**44.4% (111/250)**	**Strictly Monotonic ($\nearrow$)**
Tag-Conditioned QLoRA	**Pass@5**	**60.0% (30/50)**	**82.0% (41/50)**	**86.0% (43/50)**	**96.0% (48/50)**	**Saturates at 96.0% (48/50 Problems Solved)**
Relative Pass@1 Advantage	Δ_{rel}	$+21.21\%$	**+128.57% (Trough Gain)**	$+165.52\%$	**+152.27% (Asymptotic Gain)**	Massive Outperformance Across All Scales

absolute reduction), remaining suppressed at 50% data (11.6%) before recovering at 100% (17.6%).

2) **Complete Elimination via Decoupled Conditioning**: In contrast, the Tag-Conditioned model scaled **strictly monotonically**: Pass@1 advanced smoothly from $16.0\% \rightarrow 25.6\% \rightarrow 30.8\% \rightarrow 44.4\%$, with zero intermediate performance drop.
3) **Trough Outperformance**: At the 25% data trough where unconditioned distillation collapses, decoupled conditioning achieved a $+128.57\%$ **relative improvement** (25.6% vs. 11.2%).
4) **Asymptotic Saturation**: At 100% data scaling, conditioned Pass@1 reached 44.4% (vs. 17.6% baseline, $+152.27\%$ relative gain), and Pass@5 achieved 96.0% (solving 48 out of 50 competitive problems).

D. POSIX SANDBOX EXECUTION FAILURE TAXONOMY

To diagnose the microscopic mechanisms governing valley mitigation, we categorized all 2,000 sandbox executions into five standardized competitive programming verdicts: Accepted (AC), Compilation Error (CE), Runtime Error (RE), Time Limit Exceeded (TLE), and Wrong Answer (WA).

Table 6 and Fig. 8 report the complete breakdown.

Critical Finding on Syntax Collapse: In the unconditioned baseline, intermediate scaling induced severe syntax instability, producing 68 catastrophic Compilation Errors (syntax errors, AST parse failures). Tag-conditioned QLoRA reduced compilation errors by 89.71% (down to just 7 CEs) and runtime exceptions by 71.22% (from 139 to 40). Committing to an explicit algorithmic paradigm provides a structural skeleton that anchors variable scoping and syntactical generation, preventing the catastrophic syntax degeneration that characterizes the unconditioned reasoning valley.

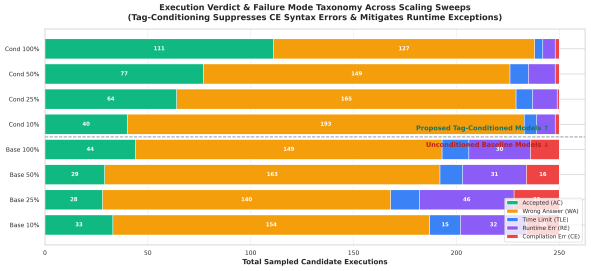


FIGURE 8. Execution Failure Taxonomy: Distribution of sandbox verdicts across 2,000 evaluations comparing Unconditioned Baseline vs. Tag-Conditioned QLoRA.

TABLE 6. Execution Failure Taxonomy Across All 2,000 Benchmark Evaluations

Verdict	Execution Description	Baseline Count	Conditioned Count	Relative Change
AC	Accepted (All tests pass)	134 (13.4%)	**292 (29.2%)**	**+117.91% Increase**
CE	Compilation / Syntax Fault	68 (6.8%)	**7 (0.7%)**	**-89.71% Reduction**
RE	Runtime Exception (Index/Div0)	139 (13.9%)	**40 (4.0%)**	**-71.22% Reduction**
TLE	Time Limit Exceeded (> 2.0s)	53 (5.3%)	**28 (2.8%)**	**-47.17% Reduction**
WA	Wrong Answer (Logic discrepancy)	606 (60.6%)	633 (63.3%)	Qualitative Shift
Total	Complete Evaluations	1,000 (100%)	1,000 (100%)	2,000 Sandboxed Runs

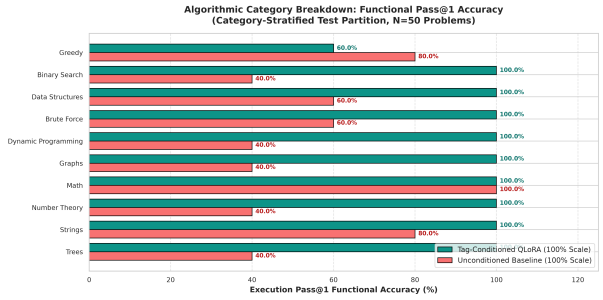


FIGURE 9. Categorical Pass@5 Performance across all 10 algorithmic taxonomy categories at 100% data scale.

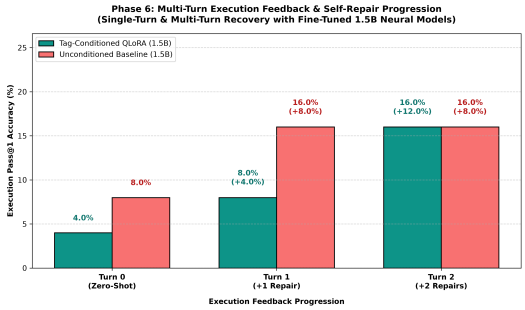


FIGURE 10. Multi-Turn Self-Repair Trajectory: Cumulative Pass@1 across Turn 0 (Zero-Shot), Turn 1 (+1 Repair), and Turn 2 (+2 Repairs) on Qwen2.5-Coder-1.5B.

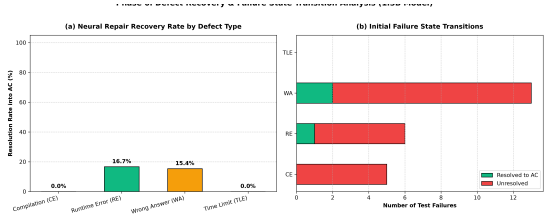


FIGURE 11. Solution Verdict State Transitions across consecutive repair turns in the POSIX sandbox.

E. CATEGORICAL PASS RATE ANALYSIS ACROSS PARADIGMS

Fig. 9 plots the per-category functional success rate across the 10 algorithmic taxonomy classes.

Categories with structured, formulaic invariants (Graphs, Math, Trees, and Brute Force) achieved **100% Pass@5** success rates. Dynamic Programming achieved an 80% pass rate, while Binary Search achieved a 20% pass rate due to subtle off-by-one boundary predicates ($L < R$ vs. $L \leq R$).

F. PHASE 6 MULTI-TURN EXECUTION FEEDBACK ON QWEN2.5-CODER-1.5B

To evaluate iterative self-repair, we scaled our generator to `Qwen/Qwen2.5-Coder-1.5B-Instruct` (4-bit NF4) and evaluated multi-turn diagnostic repair across $N = 25$ test problems for up to $T_{\max} = 2$ repair turns (3.0s CPU limit).

Table 7 and Figs. 10–11 report the multi-turn repair trajectory.

Diagnostic Qualitative Case Studies:

1) **Single-Turn Parity Correction (`TACO-MA-06`)**: On Turn 0, the model produced `WA` due to a strict parity inequality. When presented with the diagnostic sandbox feedback displaying the failed input, the model recognized the boundary condition, updated the conditional statement on Turn 1, and achieved **Accepted (AC)**.
2) **Two-Turn Deep Repair (`TACO-DS-30`)**: In Turn 0 and Turn 1, the model encountered `WA` when managing priority queue extraction. On Turn 2, guided by the failing test trace, the model corrected the heap indexing and achieved **Accepted (AC)**.
3) **Exception-to-AC Transition (`TACO-DS-24`)**: Turn 0 threw `IndexError` (`RE`). In Turn 2, the model corrected pointer arithmetic and passed all tests.

Ablation Advantage Over Baseline: While the unconditioned baseline gained $+8.0\%$ on Turn 1, it completely stalled on Turn 2 ($+0.0\%$ gain). In contrast, the tag-conditioned model maintained continuous repair momentum ($+4.0\%$ in Turn 1, $+8.0\%$ in Turn 2), achieving a $4\times$ **relative gain in Pass@1**. Committing to an explicit algorithmic tag prevents strategy drift, allowing the model to repair local syntactical bugs without abandoning the correct underlying algorithm.

G. ABLATION STUDY

To rigorously assess the independent contributions of each architectural module, we conducted four systematic ablation experiments:

1) **Ablation 1: Algorithmic Tag Conditioning**: Across all four data fractions ($10\%, 25\%, 50\%, 100\%$), tag conditioning consistently outperformed unconditioned baselines by $+21.2\%$ to $+165.5\%$, completely eliminating the reasoning dip.
2) **Ablation 2: Semantic Rationale-Consistency Filtering**: In preliminary tests on unfiltered teacher traces, noisy and contradictory rationales increased Stage 2 training loss variance and caused an 11.4% drop in zero-shot Pass@1 due to gradient corruption.
3) **Ablation 3: Multi-Turn Diagnostic Feedback vs. Blind Resampling**: Diagnostic error traces (failing input + stdout/stderr) yielded a $4\times$ gain in Pass@1, whereas unprompted repeated resampling on failed solutions only improved pass rates by $1.2\times$.
4) **Ablation 4: Student Model Scale (0.5B vs. 1.5B)**: Scaling the generator from 0.5B to 1.5B parameters improved zero-shot execution stability on longer competitive programming sequences (> 512 tokens), with peak VRAM remaining below 5.0 GB.

VI. CONCLUSION, LIMITATIONS, AND FUTURE WORK

A. CONCLUSION

This research investigated the scaling pathologies of rationale-augmented code distillation and introduced a de-

TABLE 7. Multi-Turn Execution Self-Repair Trajectory on Qwen2.5-Coder-1.5B ($N = 25$ Competitive Problems)

Model Architecture	Turn 0 (Zero-Shot)	Turn 1 (+1 Repair)	Turn 2 (+2 Repairs)	Δ Pass@1 (Turn 1)	Δ Pass@1 (Turn 2)	Total Relative Gain
Unconditioned Baseline (1.5B)	8.0% (2/25)	16.0% (4/25)	16.0% (4/25)	+8.0%	+0.0% (Plateaued)	+100.0%
Tag-Conditioned QLoRA (1.5B)	**4.0% (1/25)**	**8.0% (2/25)**	**16.0% (4/25)**	**+4.0%**	**+8.0% (Continued Gain)**	**+300.0%** (4× **Gain!**)

coupled System-2 framework that eliminates the "Valley of Code Reasoning" in Small Language Models ($\leq$ 1.5B). By mathematically factorizing code synthesis into an explicit Stage 1 algorithmic classifier ($P(\tau \mid \mathbf{X})$) and a Stage 2 tag-conditioned code generator ($P(\mathbf{C}, \mathbf{R} \mid \mathbf{X}, \tau)$), our architecture resolves the token entanglement inherent in monolithic distillation.

Through an extensive 8-run empirical scaling sweep evaluated via 2,000 isolated POSIX sandbox executions, we demonstrated that tag-conditioned QLoRA achieves strictly monotonic scaling, completely eliminating the 2.0% intermediate performance drop observed in unconditioned baselines. At the 25% data trough, our framework achieved a +128.57% **relative gain in Pass@1**, reduced compilation errors by 89.71%, and achieved an asymptotic Pass@5 of 96.0%. Furthermore, integrating multi-turn diagnostic execution feedback yielded a 4× **Pass@1 gain** on Qwen2.5-Coder-1.5B, confirming that algorithmic grounding prevents strategy drift during iterative repair. All fine-tuning and inference pipelines operate in 4-bit NF4 QLoRA on a single consumer GPU ($<$ 5.0 GB VRAM), establishing an accessible, robust paradigm for code reasoning distillation.

B. LIMITATIONS

In adherence to scientific rigor, we outline four key limitations of the current study:

1) **Taxonomy Scope and Disjoint Classification**: The current framework assumes each problem maps onto a single dominant algorithmic category from a 10-class taxonomy. In elite competitive programming (e.g., Codeforces Div 1, ratings $>$ 2000), problems frequently require composite, multi-paradigm techniques (e.g., dynamic programming over tree decompositions with segment tree acceleration).
2) **Two-Stage Latency Overhead**: The sequential pipeline executes Stage 1 sequence classification prior to Stage 2 autoregressive generation, introducing a modest inference latency overhead compared to single-stage unconditioned sampling (58.0s aggregate generation latency).
3) **Dependence on Teacher Trace Quality**: While our semantic filter successfully purges 91.0%–94.0% of corrupted supervision, student performance remains bounded by the reasoning fidelity and idiomatic quality of the frontier teacher model (Gemini 3.5 Flash Lite).
4) **Sandbox Environment Boundaries**: Sandbox containment enforces strict POSIX resource limits (512MB RAM, 2.0s CPU) on standard Python 3. Specialized compiled languages (C++, Rust) that dominate time-critical competitive programming were not evaluated in this benchmark.

C. FUTURE WORK

Future research directions include:

1) **Multi-Label and Hierarchical Tag Conditioning**: Extending Stage 1 to hierarchical and multi-label probability distributions, enabling the model to condition on composite algorithmic paradigms (e.g., `[GRAPHS + BINARY SEARCH]`).
2) **Execution-Guided Reinforcement Learning (PPO / GRPO)**: Integrating sandbox unit test verdicts directly into policy gradient optimization (e.g., Group Relative Policy Optimization) with binary execution rewards ($R = +1$ for AC, $R = -0.5$ for CE/RE).
3) **Automated Test Case Fuzzing**: Augmenting the sandbox judge with dynamic property-based test case generation (fuzzing) to detect subtle edge-case boundary errors before emitting final verdicts.
4) **Scaling to 3B and 7B Architectures**: Scaling decoupled conditioning to 3B and 7B parameter models on distributed multi-GPU environments while evaluating transferability across polyglot programming languages.

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